

COUNTEREXAMPLES TO BERENSTEIN'S CONJECTURE IN DIMENSIONS FOUR AND FOURTEEN

JIZHOU GUO

ABSTRACT. There exists a bounded strictly convex domain in $\mathbb{R}^4$, different from a ball, with real-analytic boundary diffeomorphic to S^3 , carrying a real-analytic sign-changing solution of $\Delta u + u = 0$ with $u = 0$ and $\partial_\nu u = 1$ on the boundary. There also exists a bounded convex non-ball domain in $\mathbb{R}^{14}$, invariant under the adjoint action of G_2 and with strictly convex Cartan section, carrying a sign-changing solution with zero Dirichlet data and nonzero constant normal derivative. The constructions combine planar reductions, a conformal formulation on the unit disc and interval bounds for an infinite-dimensional contraction.

1. INTRODUCTION

Let ν denote the outward unit normal. We consider the overdetermined Dirichlet eigenvalue problem

$$(1.1) \quad \Delta u + \lambda u = 0 \quad \text{in } \Omega, \quad u = 0, \quad \partial_\nu u = c \quad \text{on } \partial\Omega, \quad \lambda > 0, \quad c \neq 0.$$

Berenstein's conjecture asks whether a bounded smooth domain admitting such a solution must be a ball. The inverse spectral background is described by Berenstein [1]; related questions for the Neumann problem were studied by Berenstein–Yang [2]. A dilation normalizes λ to one, and multiplication of u by a constant normalizes c to one. For a sign-definite eigenfunction, Serrin's symmetry theorem forces the domain to be a ball [13]. The functions constructed below change sign.

The complementary Schiffer problem prescribes constant Dirichlet data and zero normal derivative for a nonconstant Helmholtz solution. The Pompeiu problem asks whether a continuous function whose integral over every rigid motion of a domain vanishes must be zero. Its Fourier-spectral formulation was developed by Brown, Schreiber and Taylor [3]. For (1.1), Green's identity with $e^{i\omega \cdot x}$, $|\omega|^2 = \lambda$, gives $\int_{\partial\Omega} e^{i\omega \cdot x} d\sigma = 0$. The boundary measure appears here, whereas the area measure appears in the corresponding Schiffer formulation.

Colbrook and Stepaniants [5] developed a conformal fixed-disc construction for the planar Schiffer and Pompeiu problems. Colbrook, Sadeghi and Stepaniants [4] adapted the method to (1.1), obtaining a bounded simply connected planar non-disc with analytic boundary and a sign-changing eigenfunction. The author's paper [8], version 2, contains constructions for the Schiffer and Pompeiu problems and a separate part giving a strictly convex planar domain for Berenstein's problem. Question 2 of [8] asks whether Berenstein's conjecture fails in every dimension $n \geq 3$, and within the convex class in those dimensions. The two constructions in the present paper address that question in dimensions four and fourteen. Dai, Sun, Wei and Zhang [6] show that a bounded uniformly convex planar domain with connected $C^{2,\epsilon}$ boundary, $\epsilon \in (0, 1)$, is a disc if (1.1) admits a nontrivial solution corresponding to a large eigenvalue.

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There are also bifurcation results for related sign-changing problems. Ruiz [12] constructs a bounded non-ball domain for a suitable C^2 semilinear nonlinearity. Dai and Zhang [7], Theorem 1.3, obtain a branch of bounded-domain solutions of $-\rho\Delta u = u - (u^+)^3$ from a bifurcation interval, for symmetry groups satisfying their spectral condition. Minlend [11], Theorem 1.1, constructs linear Dirichlet examples on unbounded periodic domains bifurcating from generalized cylinders. These results have different nonlinearities or different domain classes from the bounded linear problem considered here.

Theorem 1.1. *There exist a bounded strictly convex domain $\Omega \subset \mathbb{R}^4$, different from a ball, with real-analytic boundary diffeomorphic to S^3 , and a real-analytic sign-changing function u , analytic on a neighbourhood of $\overline{\Omega}$, such that*

$$\Delta u + u = 0 \quad \text{in } \Omega, \quad u = 0, \quad \partial_\nu u = 1 \quad \text{on } \partial\Omega.$$

Theorem 1.2. *There exists a bounded convex domain $\Omega \subset \mathbb{R}^{14}$, different from a ball, carrying a sign-changing solution of $\Delta u + u = 0$ with $u = 0$ and nonzero constant normal derivative on its boundary. After an isometric identification $\mathbb{R}^{14} \simeq \mathfrak{g}_2$, the domain is invariant under the adjoint action of the compact group G_2 , and its Cartan section is strictly convex.*

We give the planar reductions and coefficient estimates before specifying the two contraction balls. Every numerical upper bound used below is either a rational consequence of those estimates or an interval comparison in the accompanying verification files. The analytic arguments explain why the finite calculations control every infinite column and why an exact coefficient zero gives the stated boundary data.

2. THE PLANAR REDUCTIONS

2.1. Dimension four. Write $X = (x, X') \in \mathbb{R} \times \mathbb{R}^3$ and $y = |X'|$. For an $O(3)$ -invariant function $u(x, y)$, direct differentiation gives

$$(2.1) \quad \Delta_{x,y}(yu) = y \left(u_{xx} + u_{yy} + \frac{2}{y}u_y \right) = y\Delta_{\mathbb{R}^4}u, \quad y > 0.$$

Thus we seek a plane domain D , symmetric under both coordinate reflections, and an analytic function F satisfying

$$(2.2) \quad (\Delta + 1)F = 0, \quad F|_{\partial D} = 0, \quad \partial_\nu F = y \quad \text{on } \partial D, \quad F(x, -y) = -F(x, y).$$

Odd analytic division gives $F(x, y) = yG(x, y^2)$ near the axis. Consequently $u(x, X') = G(x, |X'|^2)$ is analytic across it. On the regular boundary, the product rule and $u = 0$ give $\partial_\nu F = y\partial_\nu u$. Equation (2.1) and continuity then give the equation and constant normal derivative also at the axis.

2.2. Dimension fourteen. Equip the compact Lie algebra $\mathfrak{k} = \mathfrak{g}_2$ with an $\text{Ad}(G_2)$ -invariant Euclidean inner product, and let $\mathfrak{t}$ be its two-dimensional Cartan subalgebra. Choose orthonormal coordinates $z = x + iy$ so that the reflection walls are $\arg z = k\pi/6$. The product of the six positive roots is a nonzero scalar multiple of

$$\pi(z) = \text{Im } z^6.$$

We use this normalization of π . It is harmonic. Equivalently, $\Delta\pi$ is an anti-invariant polynomial of degree four, while every anti-invariant polynomial is divisible by the six wall factors.

On the regular Cartan plane, each root plane has real dimension two. The orbit-volume density is therefore a constant times π^2 . For an invariant function u , the radial part of the Euclidean Laplacian is

$$(2.3) \quad (\Delta_{\mathfrak{k}}u)|_{\mathfrak{t}} = \pi^{-2} \text{div}_{\mathfrak{t}}(\pi^2 \nabla u) = \pi^{-1} \Delta_{\mathfrak{t}}(\pi u).$$

The second equality follows by expanding the derivatives and using $\Delta\pi = 0$. This also follows directly from the orthogonal root-plane coordinates and their Jacobian. An invariant boundary has its normal in the Cartan plane at a regular point: its tangent space contains the orbit tangent space, whose orthogonal complement is $\mathfrak{t}$. Hence, if $F = \pi u$ and $u = 0$, then

$$(2.4) \quad \partial_\nu F = \pi \partial_\nu u.$$

It suffices to construct a strictly convex Weyl-invariant section D and an anti-invariant planar field with $(\Delta + 1)F = 0$, $F = 0$ and $\partial_\nu F = \pi$ on ∂D . The lift and its singular points are treated in Section 7.

3. THE FIXED DISC AND COEFFICIENT SPACES

We use $m = 1$ for the meridian problem in dimension four and $m = 6$ for the Cartan problem in dimension fourteen. This use of $m = 1$ is a notation for the Fourier sector, rather than a Lie-algebra root count. Let

$$\psi(w) = \sum_{j \equiv 1 \pmod{2m}} c_j w^j, \quad c_j \in \mathbb{R}, \quad c_1 > 0, \quad D = \psi(\mathbb{D}).$$

Take a fixed positive field scale B , with $B = 1$ in dimension four, and put $F = BV \circ \psi^{-1}$ and $p = \psi'$. The pulled-back equation is $\Delta V + |p|^2 V = 0$.

3.1. The full zero-Dirichlet inverse. For $n \equiv m \pmod{2m}$, $n \geq m$, and $s \geq 0$, set

$$S_{n,s}(re^{i\theta}) = r^n P_s^{(0,n)}(2r^2 - 1) \sin(n\theta).$$

The Jacobi polynomials are normalized by $P_s^{(0,n)}(1) = 1$. The zero-Dirichlet inverse of Δ on this sector is

$$(3.1) \quad K_D S_{n,0} = \frac{S_{n,1} - S_{n,0}}{4(n+1)(n+2)},$$

$$(3.2) \quad K_D S_{n,s} = \frac{S_{n,s-1}}{4d(d+1)} - \frac{S_{n,s}}{2d(d+2)} + \frac{S_{n,s+1}}{4(d+1)(d+2)}, \quad d = n + 2s, \quad s \geq 1.$$

Substitution in the Jacobi differential equation verifies $\Delta K_D S_{n,s} = S_{n,s}$. The coefficients of each column sum to zero, which gives its zero boundary value. Differentiating at $r = 1$ gives

$$(3.3) \quad Ng := \partial_r K_D g|_{\mathbb{T}} = \sum_n \frac{g_{n,0}}{2(n+1)} \sin(n\theta).$$

In particular the harmonic source row $s = 0$ is retained. Removing it would remove the required nonzero normal trace. These full Dirichlet formulas are also used in [4].

Define

$$\chi_n(\theta) = \frac{\sin(n\theta)}{\sin(m\theta)} = U_{n/m-1}(\cos(m\theta)), \quad q = \frac{Ng}{\sin(m\theta)}, \quad h = \frac{\text{Im}(\psi^m)}{B \sin(m\theta)}.$$

The quotients extend analytically to the walls. With $V = K_D g$, the exact coefficient equations are

$$(3.4) \quad \mathcal{F}_1(g, c) = g + |p|^2 K_D g, \quad \mathcal{F}_2(g, c) = \frac{q^2 - h^2 |p|^2}{\eta},$$

where $\eta = 1$ in dimension four and $\eta = b$ in dimension fourteen; b is the fixed coefficient c_1 at the inverse centre. Thus η does not vary with the unknown. The boundary polynomial has degree $2m + 2$. If $q, h > 0$ and $p \neq 0$ on the boundary, a zero of (3.4) gives $q = h|p|$, which is the signed normal condition.

3.2. Norms and multiplication. For $\rho > 1$ define

$$(3.5) \quad \|g\|_G = \sum_{n,s} \rho^n |g_{n,s}|, \quad \|v\|_U = \sum_n \rho^{n-m} |v_n|, \quad \|v\|_{U,1} = \sum_n \frac{n}{m} \rho^{n-m} |v_n|,$$

$$(3.6) \quad \|c\|_C = \sigma \sum_j (j+2m) \rho^{j-1} |c_j|, \quad X = G \oplus C, \quad \mathcal{Y} = G \oplus U.$$

Both direct sums have the sum norm. We take $(\rho, \sigma) = (11/10, b^3)$ in dimension four and $(51/50, 1)$ in dimension fourteen.

Here are the multiplication estimates used throughout. In the complex mother space let $\Phi_{k,s} = r^{|k|} P_s^{(0,|k|)} (2r^2 - 1) e^{ik\theta}$, with weight $\rho^{|k|}$. For $k \geq 0$,

$$w\Phi_{k,s} = \frac{k+s+1}{k+2s+1} \Phi_{k+1,s} + \frac{s}{k+2s+1} \Phi_{k+1,s-1}.$$

For $k = -n < 0$ the corresponding coefficients are $(n+s)/(n+2s+1)$ at $(k+1, s)$ and $(s+1)/(n+2s+1)$ at $(k+1, s+1)$. The conjugate recurrence follows by conjugation. All coefficients are nonnegative and sum to one. Iteration therefore gives

$$(3.7) \quad \|af\|_G \leq \|a\|_W \|f\|_G, \quad \|a\|_W = \sum_k \rho^{|k|} |a_k|.$$

The sine embedding uses conjugate coefficients of magnitude $|g_{n,s}|/2$, so its mother-space norm is exactly $\|g\|_G$.

The character identity $U_a U_b = \sum_{k=|a-b|}^{a+b} U_k$ has $\min(a, b) + 1$ terms. The weights of its terms are bounded by the product weight. It follows that

$$(3.8) \quad \|v_1 v_2\|_U \leq \min\{\|v_1\|_{U,1} \|v_2\|_U, \|v_1\|_U \|v_2\|_{U,1}\}.$$

An even Laurent multiplier with frequencies in $2m\mathbb{Z}$ has U -operator norm at most its Wiener norm: use the two-character expression for a paired cosine. In particular multiplication by $|p|^2$ costs at most P^2 , where $P = \sum_j j \rho^{j-1} |c_j|$.

The absolute column sums in (3.1)–(3.2) give

$$(3.9) \quad \|K_D\|_{G \rightarrow G} \leq \kappa_m := \max \left\{ \frac{1}{2(m+1)(m+2)}, \frac{1}{(m+2)(m+4)} \right\}.$$

Thus $\kappa_1 = 1/12$ and $\kappa_6 = 1/80$. From (3.3) and the shape weights,

$$(3.10) \quad \|q(g)\|_U \leq \frac{\rho^{-m}}{2(m+1)} \|g\|_G, \quad \|q(g)\|_{U,1} \leq \frac{\rho^{-m}}{2m} \|g\|_G,$$

$$(3.11) \quad \sum_j \rho^{j-1} |\delta c_j| \leq \frac{\|\delta c\|_C}{\sigma(1+2m)}, \quad \sum_j j \rho^{j-1} |\delta c_j| \leq \frac{\|\delta c\|_C}{\sigma}.$$

These estimates and finite character fusion prove that (3.4) is a continuous polynomial $X \rightarrow \mathcal{Y}$.

No exponential radial weight is imposed on s . This causes no regularity assumption on the solution: $|S_{n,s}| \leq 1$ on the disc, so source truncations converge uniformly. The zero-Dirichlet $W^{2,a}$ estimates for any finite $a > 2$ give C^1 convergence of their solutions and convergence of the normal traces. Consequently $V = K_D g$ has the stated boundary values and solves $\Delta V = g$ weakly. At an exact zero the elliptic equation gives interior analyticity. Analyticity across the boundary is addressed below.

4. A CONTRACTION CRITERION

Let $A : \mathcal{Y} \rightarrow X$ be a bounded injective operator. Let x^0 be the centre used to construct A and let $\bar{x}$ be the centre of the contraction ball. Write

$$Y \geq \|A\mathcal{F}(\bar{x})\|, \quad Z_0 \geq \|I - ADF(x^0)\|, \quad Z_1 \geq \|A(DF(\bar{x}) - DF(x^0))\|.$$

Suppose that expanding the polynomial at $\bar{x}$ gives multilinear remainder bounds d_q , $2 \leq q \leq 2m+2$. Set $C_q = \|A\| d_q$ and

$$R(r) = \sum_{q=2}^{2m+2} C_q r^q, \quad Z_2(r) = \sum_{q=2}^{2m+2} q C_q r^{q-1}.$$

Thus Z_2 denotes the nonlinear derivative contribution at radius r . The inequalities

$$(4.1) \quad Y + (Z_0 + Z_1 - 1)r + R(r) < 0, \quad Z_0 + Z_1 + Z_2(r) < 1$$

imply existence of a unique zero in $\overline{B}_X(\bar{x}, r)$. Indeed $x \mapsto x - A\mathcal{F}(x)$ maps this complete closed ball into itself and is a strict contraction. Its fixed point satisfies $A\mathcal{F}(x) = 0$; injectivity of A gives $\mathcal{F}(x) = 0$. The factor q in Z_2 is obtained by replacing one of the q multilinear factors by a direction vector.

5. THE CERTIFICATE IN DIMENSION FOUR

5.1. Finite centre and approximate inverse. The centre $\bar{x} = x^0$ is given by `finite_center_r4_M31_S24.json`. Its hexadecimal coefficients are exact dyadic rationals. The source coordinates are $n = 1, 3, \dots, 31$, $s = 0, \dots, 24$, and the shape coordinates are $j = 1, 3, \dots, 31$. There are $400 + 16 = 416$ finite unknowns. For orientation, $b = c_1 = 7.08373029362\dots$, $c_3 = 0.153114245\dots$ and $c_5 = 0.171014117\dots$

Let P_f and P_t project onto these finite coordinates and their complements, in the appropriate source and boundary spaces. The finite Jacobian $J_f = P_f DF(\bar{x})|_{X_f}$ is assembled from four disjoint interval column files. The verifier computes an interval inverse, takes its exact dyadic midpoint A_f , and recomputes its defect. Every midpoint entry is compared with the saved inverse. The resulting bounds are

$$(5.1) \quad \|A_f\| < 1665.434, \quad E_f := \|I - A_f J_f\| < 2.05 \cdot 10^{-34} < 21 \cdot 10^{-35}.$$

The X -norm of the last column of A_f , without the input row weight, is < 54.778 . These are direct interval comparisons; the additional `check_inequalities.py` also checks the sharper defect bound in (5.1). Thus A_f and J_f are invertible.

On high shape indices define the ball operator

$$(5.2) \quad (Tc)_j = -b^3(j+2)(c_j - c_{j+2}), \quad (T^{-1}r)_j = - \sum_{\substack{k \geq j \\ k \text{ odd}}} \frac{r_k}{b^3(k+2)}, \quad j \geq 33.$$

The ball shape column has coefficients $-b^3(j+2)$ at character row j and $b^3 j$ at row $j-2$, by differentiating (3.4) at $\psi = bw$, $h = b$, $q = b^2$. Weighted summation of (5.2) gives

$$\|T^{-1}\| \leq L_T = (1 - \rho^{-2})^{-1}.$$

For $y = (f, v) \in \mathcal{Y}$, set $g_t = f_t$ and

$$(5.3) \quad c_t = T^{-1}P_t(v - 2q_0 q(f)), \quad q_0 = q(g^0).$$

Subtract $b^3 33 c_{33,t}$ from row 31 of the finite boundary output $P_f y$, and apply A_f to that corrected finite output. Together with (g_t, c_t) this defines Ay . The finite row correction is the ball coupling from shape index 33; it is part of the operator.

The operator is injective. If $Ay = 0$, its high source component gives $f_t = 0$, its high shape component gives $P_t(v - 2q_0q(f)) = 0$, and the invertible finite map gives $f_f = v_f = 0$, since the row correction is then zero. Thus $f = 0$, $v_t = 0$ and $y = 0$. Put $N_q = \|q_0\|_{U,1} < 86.059$ and $E = 54.778(33/35)\rho^{-32}$. Equations (3.10), (5.2) and the finite column bound give

$$(5.4) \quad \|A\| \leq 1665.434 + 1 + (1 + E)L_T \frac{N_q}{2\rho} < 2444.$$

The complete residual is expanded without discarding any output coefficient and preconditioned with (5.3). Its finite and tail norms are respectively $< 1.198868 \cdot 10^{-6}$ and $< 1.838234 \cdot 10^{-6}$, giving

$$(5.5) \quad Y < 3.038 \cdot 10^{-6}.$$

5.2. All derivative columns. In a weighted ℓ^1 space an operator norm is the supremum of its normalized column norms. The column partition is as follows:

Class	Indices	Upper bound
Finite	All 416 finite coordinates	$0.578 + E_f$
Near source	$n \leq 31$, $25 \leq s \leq 55$; and $33 \leq n \leq 61$, $0 \leq s \leq 55$; 1336 columns	0.679
Far source	$n \geq 63$ or $s \geq 56$	0.886
Near shape	$j = 33, 35, \dots, 299$; 134 columns	0.720
Far shape	Every odd $j \geq 301$	0.925823

The finite-tail, near-source and near-shape files enclose the whole output of (5.3), including the finite row correction. Their spans are checked for exact coverage, and all their columns were regenerated in the complete replay.

Here are the analytic arguments for the unenumerated classes. Let $P = \sum_j j\rho^{j-1}|c_j| < 9.09$ and $d = |p_0|^2 K_D e_{n,s}$ for a source column. Its source derivative is $e_{n,s} + d$, and its boundary derivative is exactly $2q_0q(e_{n,s})$. The high boundary residual in (5.3) is therefore $-2P_t(q_0q(d))$. Multiplication by $|p_0|^2$ shifts angular degree by at most 30 and lowers the radial index by at most 30. K_D lowers it by at most one more. Hence $n \geq 63$ cannot reach the finite angular rows, and $s \geq 56$ cannot reach the finite radial rows. For $n \geq 63$ the trace quotient costs at most $[2\rho(n-29)]^{-1}$. Its whole-column bound is

$$(5.6) \quad \frac{P^2}{65 \cdot 67} \left(1 + (1 + E)L_T \frac{N_q}{34\rho} \right) < 0.885833.$$

For $s \geq 56$, $n \leq 61$, the harmonic row of d vanishes, and the bound is $P^2/(113 \cdot 115) < 0.006359$. These cases cover the complement of the near-source list.

For a high shape column, the pure boundary output, aligned by offsets from j , has the form

$$(5.7) \quad B_{j+a}^{\text{shape}} = jA_a + B_a, \quad a = -92, -90, \dots, 60, \quad \sum_a A_a = 0, \quad j \geq 125.$$

To verify the identity analytically, expand $-2h_0\chi_j|p_0|^2 - h_0^2(jw^{j-1}\bar{p}_0 + j\bar{w}^{j-1}p_0)$ in characters. The centre has character degree at most 30 and derivative frequency at most 30. Once the high frequency is separated from these supports, character fusion has fixed offsets and coefficients independent of j , apart from the displayed derivative factor j . The latter is a difference of adjacent characters, using $2\cos(k\theta) = U_k - U_{k-2}$. The two fused sums have equal numbers of terms, so their coefficient sums cancel. This proves the last identity in (5.7). The two exact columns $j = 125, 127$ determine A_a, B_a ; the program also checks $j = 129, 301$ exactly.

Put $D_a = B_a - (a + 2)A_a$. Formula (5.2) gives

$$c_{j+v} = -b^{-3} \sum_{a \geq v} \left(A_a + \frac{D_a}{j + a + 2} \right).$$

For offsets $v < -92$ the A_a sum is zero. Let

$$\begin{aligned} L_v &= -b^{-3} \sum_{a \geq v} A_a - \mathbf{1}_{\{v=0\}}, & L_0^\sharp &= \sum_v \rho^v |L_v|, \\ L_1^\sharp &= \sum_v |v| \rho^v |L_v|, & D^\sharp &= b^{-3} \sum_v \rho^v \sum_{a \geq v} |D_a|, \\ D^- &= \frac{\rho^{-94}}{b^3(1 - \rho^{-2})} \sum_a |D_a|. \end{aligned}$$

The sums over v here run through the offsets in (5.7). At $J = 301$, the normalized boundary defect is bounded, for every $j \geq J$, by

$$(5.8) \quad L_0^\sharp + \frac{L_1^\sharp}{J+2} + \left(1 + \frac{60}{J+2} \right) \frac{D^\sharp}{J-90} + \frac{D^-}{J-90} < 0.706765.$$

This follows by bounding $(j + v + 2)/(j + 2)$ in the finite sums and extending the low-offset leakage to its geometric series.

The interior shape output uses the zero-Dirichlet factor $V_0 = (1 - r^2)Q_0$. The finite rational identity is checked coefficient by coefficient, independently of its interval evaluation. For angular degree $n > 0$ the r^2 recurrence has coefficients

$$a_s = \frac{(s+1)(n+s+1)}{(n+2s+1)(n+2s+2)}, \quad c_s = \frac{s(n+s)}{(n+2s)(n+2s+1)}, \quad b_s = 1 - a_s - c_s.$$

Thus $a_s, c_s \geq 0$ and $a_s + c_s \leq (2s+1)/(n+1)$. The absolute column sum of $1 - r^2$ is $2(a_s + c_s)$. Set $W = \bar{p}_0 Q_0$, expanded in the complex mother basis. Its least angular frequency is -61 . Multiplication by w^{j-1} increases the radial index by at most $\max(0, -k)$ for an input frequency k , while its coefficients remain nonnegative with sum one. The two conjugate shape terms consequently have normalized source norm

$$(5.9) \quad I_J = \frac{4}{b^3} \sum_{k,s} \frac{|W_{k,s}| \rho^k [2(s + \max(0, -k)) + 1]}{J+k} < 0.076108.$$

The additional tail shape norm from its trace is at most $L_T N_q I_J / [\rho(J-61)] < 0.142951$. The only finite output is the row correction in (5.3). The boundary and trace contributions to it are bounded by

$$\frac{54.77833 \sum_a |D_a|}{b^3(J-90)(J+2)\rho^{J-1}} + \frac{54.77833 \rho^{-(J-93)}}{J-90} \frac{N_q I_J}{\rho(J-61)} < 5.23 \cdot 10^{-10}.$$

Together with (5.8) and (5.9) this gives the far-shape bound 0.925823. Every denominator increases with $j \geq J$ and every leakage weight decreases. The estimate therefore applies to the entire tail.

The five classes now give

$$(5.10) \quad Z_0 = \|I - \text{ADF}(\bar{x})\| < 463/500 = 0.926, \quad Z_1 = 0.$$

5.3. Nonlinear bounds and radius. At the centre put $P = \|p_0\|_W$, $P_2 = \| |p_0|^2 \|_W$, $H_0 = \|h_0\|_U$, $H_1 = \|h_0\|_{U,1}$, $H_{20} = \|h_0^2\|_U$, $V_0^\# = \|K_{DG}^0\|_G$. For a unit X perturbation the linear bounds are

$$a_p = a_{h,1} = b^{-3}, \quad a_{h,0} = (3b^3)^{-1}, \quad a_{q,0} = (4\rho)^{-1}, \quad a_{q,1} = (2\rho)^{-1}, \quad K = 1/12.$$

Let $L_h = 2 \min(H_1 a_{h,0}, H_0 a_{h,1})$ and $Q_h = a_{h,0} a_{h,1}$. Expansion of (3.4), using (3.8), gives

$$\begin{aligned} d_{2,I} &= 2Pa_p K + V_0^\# a_p^2, & d_{3,I} &= K a_p^2, \\ d_{2,B} &= a_{q,0} a_{q,1} + H_{20} a_p^2 + 2Pa_p L_h + Q_h P_2, \\ d_{3,B} &= L_h a_p^2 + 2Pa_p Q_h, & d_{4,B} &= Q_h a_p^2. \end{aligned}$$

The interval computation of these expressions and (5.4) gives

$$C_2 < 268, \quad C_3 < 23/10000, \quad C_4 < 6/10^8.$$

At the exact rational radius $r = 1/12500$, the rational upper bounds in (5.5) and (5.10) give

$$(5.11) \quad Y + (Z_0 - 1)r + 268r^2 + \frac{23}{10000}r^3 + \frac{6}{10^8}r^4 < -1.1667 \cdot 10^{-6},$$

$$(5.12) \quad Z_0 - 1 + 536r + \frac{69}{10000}r^2 + \frac{24}{10^8}r^3 < -0.0311.$$

These are strict rational comparisons. Section 4 gives an exact zero x^* in this ball.

5.4. Geometric and sign margins. For a perturbation in this ball,

$$|\delta c_1| \leq \frac{r}{3b^3}, \quad \sum_{j \geq 1} j |\delta c_j| R^{j-1} \leq \frac{r}{b^3} \quad (R \leq \rho), \quad \sum_{j \geq 3} j(j-1) |\delta c_j| \leq \frac{r\rho}{b^3 e \log \rho}.$$

The last estimate uses $\sup_{t \geq 0} t\rho^{-t} = 1/(e \log \rho)$. The interval sums at the centre, with these perturbation bounds, give

$$(5.13) \quad \operatorname{Re} \psi'(w) > 5.39635 \quad (|w| \leq 21/20),$$

$$(5.14) \quad |\psi'(w)| > 5.66945, \quad \operatorname{Re} \left(1 + \frac{w\psi''(w)}{\psi'(w)} \right) > 0.10935 \quad (|w| \leq 1),$$

$$(5.15) \quad \operatorname{Re} \frac{w\psi'(w)}{\psi(w)} > 0.84029 \quad (|w| = 1), \quad |c_3^*| > 0.1531.$$

For example the convexity lower bound is $1 - (\sum_{j \geq 1} j(j-1)|c_j| + r\rho/(b^3 e \log \rho))/(b - \sum_{j \geq 1} j|c_j| - r/b^3)$. The positive real derivative gives univalence by integrating along line segments. The curvature and radial inequalities give strict convexity and strict star-shapedness.

The character bound $|U_{n-1}| \leq n$ gives $q^* > 24.345$. The same estimate for h gives $h^* \geq b - \sum_{j \geq 1} j|c_j| - r/b^3 > 5.66945$; thus its positivity is also a consequence of the derivative margin. The squared boundary equation therefore yields the positive branch $q^* = h^*[p^*]$. The interval values at $w = i/4, 3i/4$, including the uniform field error $r/12$, satisfy

$$(5.16) \quad V^*(i/4) > 10, \quad V^*(3i/4) < -8.$$

The centre values are enclosed near 13.53518938515 and -8.60629803547 , respectively. The two images lie on the positive imaginary axis, since $\operatorname{Re} \psi' > 0$. Thus division by y preserves their opposite signs.

6. THE CERTIFICATE IN DIMENSION FOURTEEN

6.1. Two centres and the finite block. All norms in this section use $\rho = 51/50$. The inverse centre x^0 is specified by `center_r14_M114_S36_arbrefined.json`, and the contraction centre $\bar{x}$ by `center_r14_M186_S48_arbrefined2.json`. They have the same exact dyadic field scale B . At x^0 the finite source indices are $n = 6, 18, \dots, 114$, $s = 0, \dots, 36$, and the finite shape indices are $j = 1, 13, \dots, 109$. There are $370 + 10 = 380$ finite unknowns. At $\bar{x}$ the corresponding cutoffs are 186 and 48.

The saved binary64 matrix A_f is interpreted entrywise as an exact dyadic matrix. Eight disjoint files enclose all finite Jacobian columns at 128-bit precision. Direct interval multiplication gives

$$(6.1) \quad \|A_f\| < 82859, \quad \|I - A_f J_f\| < 2.161 \cdot 10^{-9} < 3 \cdot 10^{-9}.$$

The program also stores and recomputes the norm of each output column of A_f , including its row weight. This profile is used for the source cross bound. Both A_f and J_f are invertible.

6.2. The tail Schur operator. At x^0 write

$$Qf = \frac{2q_0 q(f)}{b}, \quad H = D_c \mathcal{F}_1(x^0), \quad \mathcal{T} = D_c \mathcal{F}_2(x^0), \quad C = P_t(\mathcal{T} - QH)|_{C_t}, \quad b = c_1^0.$$

The projections onto the high shape and boundary coordinates are identified by boundary index $n = j + 5$. Put $h_b = b^6/B$, $\alpha = h_b^2$ and $\tau = (1 - \rho^{-12})^{-1}$. The ball shape operator has columns

$$\mathcal{T}_0 e_j = -\alpha[(j+12)\chi_{j+5} - j\chi_{j-7}].$$

Its step-12 tail inverse satisfies $\|\mathcal{T}_0^{-1}\| \leq \tau/\alpha$, by the same weighted summation as (5.2). Define $\mathcal{K} = I - \mathcal{T}_0^{-1}C$. The 20 columns $j = 121, 133, \dots, 349$ are enclosed separately and have maximum < 0.059850 . We next bound every $j \geq J = 361$ analytically.

Put $P = \sum_j j \rho^{j-1} |c_j^0|$, $C_\psi = \sum_j \rho^{j-1} |c_j^0|$, $\delta P = \sum_{j>1} j \rho^{j-1} |c_j^0|$, $\delta h_1 = \|h_0 - h_b\|_{U,1}$ and $\delta h_{20} = \|h_0^2 - \alpha\|_U$. Set

$$D_h = \frac{6C_\psi^5}{B}, \quad \delta D_h = \frac{6(C_\psi^5 - b^5)}{B}, \quad D_{h,b} = \frac{6h_b}{b}, \quad \delta P_2 = 2b \delta P + (\delta P)^2.$$

Here $C_\psi \geq b$, and the nonnegative power expansion bounds the coefficient difference from the ball by δD_h . Expansion of $-2h_0 \delta h |p_0|^2 - h_0^2 \delta |p|^2$, followed by (3.8) and division by the shape weight, bounds $\mathcal{T} - \mathcal{T}_0$ by

$$B_J = \frac{2}{b(J+12)} (\delta h_1 D_h P^2 + h_b \delta D_h P^2 + h_b D_{h,b} \delta P_2) + \frac{2}{b} (P \delta h_{20} + \alpha \delta P).$$

After $\mathcal{T}_0^{-1}$ this contributes $\tau B_J / \alpha < 0.275974$.

For the other Schur term factor $K_D g^0 = (1 - r^2)Q_0$ and set $W = \bar{p}_0 Q_0$. The factor identity is checked over $\mathbb{Q}$; its interval expansion satisfies the same identity. For a mother-space term at angular frequency k , multiplying by w^{j-1} gives frequency $j - 1 + k > 0$ for $j \geq J$. The nonnegative multiplication coefficients sum to one. In the harmonic output row of $(1 - r^2)\Phi_{n,s}$ only $s = 0, 1$ can contribute. Their coefficients have magnitude at most $1/(n+2)$; taking the normal trace costs a further $1/[2(n+1)]$. Consequently, with $N_q = \|q_0\|_{U,1}$, the QH contribution to $\mathcal{K}$ is at most

$$(6.2) \quad \frac{4\tau N_q}{\alpha b \rho^6} \sum_{k,s} \frac{|W_{k,s}| \rho^k}{(J+k)(J+k+1)} < 0.197588.$$

This retains the harmonic row, rather than replacing it by the entire K_D column sum. Both denominator factors increase with $j \geq J$. The total far bound is < 0.473562 . Combining near and far columns gives

$$(6.3) \quad \|\mathcal{K}\| < 0.474, \quad \|C^{-1}\| \leq \frac{\tau}{\alpha(1 - \|\mathcal{K}\|)} < 9.$$

This covers the whole high shape space.

We use the finite polynomial

$$(6.4) \quad C_{13}^{-1} = \left(\sum_{\ell=0}^{12} \mathcal{K}^\ell \right) \mathcal{T}_0^{-1}.$$

It has norm < 9 and is invertible: its polynomial factor satisfies $(\sum_{\ell=0}^{12} \mathcal{K}^\ell)(I - \mathcal{K}) = I - \mathcal{K}^{13}$, and both $I - \mathcal{K}$ and $I - \mathcal{K}^{13}$ are invertible since $\|\mathcal{K}\| < 1$.

6.3. The full approximate inverse and cross bounds. For $y = (f, v) \in \mathcal{Y}$, define

$$(6.5) \quad \tilde{v} = v - Qf, \quad c_t = C_{13}^{-1} P_t \tilde{v}, \quad g_t = P_t f - P_t H c_t,$$

$$(6.6) \quad x_f = A_f P_f (y - D\mathcal{F}(x^0)(g_t, c_t)), \quad Ay = (x_f, g_t, c_t).$$

The output shear, the invertible high shape map, the source shear and the finite invertible map are triangular factors. More directly, $Ay = 0$ implies $c_t = g_t = 0$, hence $P_t f = 0$ and $P_t(v - Qf) = 0$, and then $x_f = 0$ implies $P_f f = P_f v = 0$. It follows that $y = 0$. The same equations reconstruct y from any x_f, g_t, c_t , so A is also a bounded bijection.

The cross bounds are

$$(6.7) \quad X_s := \|A_f P_f D_g \mathcal{F}|_{G_t}\| < 0.171168 < 1/5, \quad X_c := \|A_f P_f D_c \mathcal{F}|_{C_t}\| < 9.183045 < 10.$$

We describe their complete support coverage. Let $p_{\max} = 108$, $h_{\max} = 6 \cdot 109 = 654$ and $h_{\max}^2 = 2h_{\max} - 6 = 1302$. For the source bound, a pair of derivative monomials $w^a \bar{w}^b$ is bounded by weight ρ^{a+b} and has angular displacement at most $a + b$, while its radial lowering is at most $a + b$. Multiplication positivity and the exact K_D column sum bound each finite output by the largest inverse-profile entry among the allowed output rows. The harmonic boundary output is expanded by character fusion. Enumerating $n \leq 330$, $s \leq 253$, outside the finite block, gives 6742 possible source columns. For every remaining column the finite source output is zero by support, and the finite boundary output is zero either because $s > 0$ or because the character supports are separated. The certified maximum in (6.7) is reached at $(n, s) = (6, 37)$.

For the shape bound the finite output is exactly zero when

$$j + 5 - h_{\max} - p_{\max} > 114, \quad j - 1 - p_{\max} - h_{\max}^2 > 114, \quad j - 1 - 114 - p_{\max} > 114.$$

All three hold for $j \geq 1537$. The 118 columns $j = 121, 133, \dots, 1525$ are expanded and multiplied by A_f with interval arithmetic. Their index lists and centre and inverse hashes are checked for complete coverage. Thus neither cross bound is a sampled tail estimate.

The source smoothing and high shape derivative bounds are

$$(6.8) \quad \|P_t|p_0|^2 K_D|_{G_t}\| < 0.141426 < 0.142, \quad \|H|_{C_t}\| < 250.$$

For the source estimate use P^2 times the largest high-source K_D column sum: compare the lowest high radial index $s = 37$ and the lowest high angular index $n = 126$, including both harmonic and nonharmonic columns. For the shape estimate the 20 near columns are enclosed separately. The far bound comes from the factor $1 - r^2$, as in (5.9), and equals

$$4 \sum_{k,s} \frac{|W_{k,s}| \rho^k [2(s + \max(0, -k)) + 1]}{361 + k} < 249.711 < 250.$$

For the feedback block, retain only those terms in (3.1) and (3.2) that can reach radial row zero after multiplication. If a multiplier pair has total degree $d = a + b$, only K_D terms at radial index at most d contribute, and their resulting angular degree is at least $\max(6, n - d)$. The normal quotient therefore has the additional factor $1/[2(\max(6, n - d) + 1)]$. Summing these nonnegative majorants proves

$$(6.9) \quad \|Q|p_0|^2 K_D|_{G_t}\| < 0.000112959 < 114/10^6.$$

The program checks 5952 cases with $n \leq 342$, $s \leq 217$, outside the finite block. For $s \geq 218$ no harmonic output is possible; for $n > 342$ both the K_D coefficients and the angular trace factors decrease. This proves the bound on the unenumerated complement.

For the norm of A on the entire output space, the coarse estimates $\|H\| < 1076$ and $\|Q\| < 0.133 < 1$ follow from $2P\|K_D g^0\|_G$ and (3.10). Equations (6.5)–(6.6) yield

$$(6.10) \quad \|A\| \leq 82859 + (1 + X_s) + ((1 + X_s)\|H\| + X_c + 1)9 < 94580 < 10^5,$$

where $X_s = 1/5$, $X_c = 10$ may be used as rational ceilings.

6.4. Every column of the derivative defect. For a normalized high source input, the transformed boundary output in (6.5) is $-Q|p_0|^2 K_D e$. Its shape correction has norm at most $c = 9(114/10^6) = 513/500000$. The source error has norm at most $g = 0.142 + 250c = 797/2000$. The finite correction is bounded by $X_s g + X_c c$. Thus the whole column defect is bounded by

$$(6.11) \quad (1 + X_s)g + (1 + X_c)c = 244743/500000 = 0.489486 < 0.49.$$

For a high shape input, (6.4) gives the exact shape remainder $\mathcal{K}^{13}e$. Its source remainder is $H\mathcal{K}^{13}e$, and its finite remainder is controlled by the two cross blocks. Consequently every such column has defect at most

$$(6.12) \quad ((1 + X_s)250 + 1 + X_c)(0.474)^{13} < 0.018962.$$

For a finite normalized input, let $(f, v) = D\mathcal{F}(x^0)e$ and set $c_0 = \mathcal{T}_0^{-1}P_t(v - Qf)$. The same algebra gives the coarse bound

$$(6.13) \quad E_f + \frac{6}{5}\|P_t f\| + \frac{311}{1 - 0.474}\|c_0\|, \quad E_f = 3 \cdot 10^{-9}.$$

Eight interval files cover all 380 finite input columns. For indices $0, \dots, 378$, (6.13) is < 0.114825 . The final shape column, index 379, has a larger coarse estimate and is treated directly.

For that column let $\hat{c}$ be the eight-term dyadic shape vector in `r14_M114_lastshape_tail_seed.json` and put $R = P_t(v - Qf) - C\hat{c}$. Full coefficient expansion gives $\|R\| < 3.573 \cdot 10^{-15}$. The difference between $C_{13}^{-1}P_t(v - Qf)$ and $\hat{c}$ is at most

$$\epsilon_c = 9\|R\| + \frac{0.474^{13}}{1 - 0.474}\|c_0\|.$$

The second term compares the finite polynomial with the true Schur inverse. Let $\hat{g} = P_t f - P_t H \hat{c}$. Expand its complete finite correction by (6.6). The sum of its finite, source-tail and shape-tail defects, plus $311\epsilon_c$ divided by the input weight, is < 0.160581 . Thus index 379 is included with a rigorous remainder, rather than accepting its finite seed as an exact solution.

Finite, high source and high shape inputs exhaust X . The preceding estimates establish

$$(6.14) \quad Z_0 = \|I - AD\mathcal{F}(x^0)\| < 0.49.$$

6.5. Nonlinear bounds and centre transfer. For a unit shape perturbation,

$$\theta_0 = 1/13, \quad \theta_1 = \sup_{j \equiv 1 \pmod{12}} \frac{j}{j+12} = 1$$

bound $\sum \rho^{j-1} |\delta c_j|$ and $\sum j \rho^{j-1} |\delta c_j|$, respectively. Both estimates use the weighted norms in (3.11).

At either centre let $H_{0,0} = \|h\|_U$, $H_{1,0} = \|h\|_{U,1}$. For $1 \leq k \leq 6$ define

$$H_{0,k} = \frac{\binom{6}{k}}{B} C_\psi^{6-k} \theta_0^k,$$

$$H_{1,k} = \frac{\binom{6}{k}}{B} \left(\frac{6-k}{6} P C_\psi^{5-k} \theta_0^k + \frac{k}{6} C_\psi^{6-k} \theta_1 \theta_0^{k-1} \right).$$

The term with $6-k$ is zero when $k=6$. These follow by expanding ψ^6 and differentiating a product of six holomorphic factors; the character multiplier $n/6$ accounts for the total degree divided by six. Put

$$L_\ell = \sum_{\substack{0 \leq k, k' \leq 6 \\ k+k'=\ell}} \min(H_{1,k} H_{0,k'}, H_{0,k} H_{1,k'}), \quad a_{q,0} = \rho^{-6}/14, \quad a_{q,1} = \rho^{-6}/12.$$

Take $L_\ell = 0$ outside $0 \leq \ell \leq 12$. The boundary majorant at degree q is

$$d_{q,B} = \frac{1}{c_1} (P^2 L_q + 2P\theta_1 L_{q-1} + \theta_1^2 L_{q-2} + \mathbf{1}_{\{q=2\}} a_{q,0} a_{q,1}).$$

For the interior part, $d_{2,I} = 2P\theta_1 \kappa_6 + \|K_{Dg}\|_G \theta_1^2$, $d_{3,I} = \kappa_6 \theta_1^2$, and the other interior coefficients are zero. Set $d_q = d_{q,I} + d_{q,B}$, $2 \leq q \leq 14$. These are multilinear bounds in the sum input norm.

At x^0 , $\sum_{q=2}^{14} q d_q < 37.149$. The complete weighted distance between the centres is $\delta = \|\bar{x} - x^0\| < 6.440 \cdot 10^{-18} < 1$. Therefore

$$(6.15) \quad \begin{aligned} Z_1 &\leq \|A\| \sum_{q=2}^{14} q d_q \delta^{q-1} \\ &< 10^5 \cdot 37.149 \cdot 6.440 \cdot 10^{-18} < 2.4 \cdot 10^{-11}, \\ Z_0 + Z_1 &< 0.490000000024 < 1/2. \end{aligned}$$

This uses $\delta^{q-1} \leq \delta$ for $q \geq 2$.

At $\bar{x}$ the complete coefficient residual is $< 1.063 \cdot 10^{-30}$. The evaluator uses the denominator $\bar{c}_1$ in the boundary output, whereas (3.4) fixes $b = c_1^0$. Multiply its residual and nonlinear coefficients by $f = \max(1, \bar{c}_1/b) < 1 + 1.7 \cdot 10^{-23}$ to bound the fixed map. Both nonlinear certificates use $\theta_1 = 1$, and their quadratic coefficients satisfy $d_2 < 18.551$. Hence $Y < 1.063 \cdot 10^{-25}$.

At $r = 10^{-20}$, using 10^5 and $1/2$ as ceilings for the inverse norm and transferred defect, respectively, the interval comparisons give

$$(6.16) \quad Y - \frac{r}{2} + \sum_{q=2}^{14} 10^5 f d_q r^q < -4.99989 \cdot 10^{-21},$$

$$(6.17) \quad -\frac{1}{2} + \sum_{q=2}^{14} q 10^5 f d_q r^{q-1} < -0.499999999999996.$$

In particular $Z_2(r) < 4 \cdot 10^{-14}$. The exact zero x^* follows from Section 4.

6.6. Geometry and sign. At $\bar{x}$ put $L = \bar{c}_1 - \sum_{j>1} j|\bar{c}_j|$ and $U = \sum_{j>1} j(j-1)|\bar{c}_j|$. The radius bounds $|\delta\psi'| \leq r$ on $|w| \leq 1.001$ and $|\delta\psi''| \leq 50r$ on $|w| \leq 1$ follow from (3.11) and $1/(e \log(51/50)) < 50$. The centre sums certify a positive outer derivative lower bound and $L - U - 51r > 25.4946 > 0$. Hence ψ^* is univalent on $|w| \leq 1.001$, and $\operatorname{Re}(1 + w\psi''/\psi') > 0$ on the unit disc. Its image is strictly convex and contains zero. The interval comparison $|\bar{c}_{13}| - r/[25\rho^{12}] > 0$ proves $c_{13}^* \neq 0$.

For a character polynomial, its constant coefficient minus $\sum_{n>6} (n/6)|v_n|$ is a lower bound on its boundary value. The source perturbation costs at most $r\rho^{-6}/12$, and the shape perturbation costs at most $\sum_{k=1}^6 H_{1,k}r^k$. The exact-boundary margins are

$$(6.18) \quad q^* > 29.1875, \quad h^* > 0.98405.$$

Thus the positive square root is obtained throughout the boundary. At $w = te^{i\pi/12}$ the centre values for $t = 1/10, 37/100$ are enclosed below -0.0795470 and above 1.4919379 , respectively. The field perturbation is at most $r/80$, so the exact values still have opposite signs. Real coefficients $j \equiv 1 \pmod{12}$ keep this ray fixed, and its image has positive radius because $\operatorname{Re} \psi' > 0$. Therefore $\pi(\psi(w)) > 0$ at both points.

7. RECONSTRUCTION OF THE DOMAINS AND FIELDS

7.1. The planar field and exact boundary data. At either exact zero, $V = K_D g$ has zero Dirichlet trace and satisfies $\Delta V + |p|^2 V = 0$. The conformal map is univalent on a neighbourhood of the closed disc. Set $F = BV \circ \psi^{-1}$ on $D = \psi(\mathbb{D})$. The conformal normal derivative formula gives

$$\partial_\nu F(\psi(e^{i\theta})) = B \frac{Ng(\theta)}{|p(e^{i\theta})|}.$$

The positive margins for q, h give $Ng = h|p|\sin(m\theta)$. Thus $\partial_\nu F = \operatorname{Im} \psi$ for $m = 1$ and $\partial_\nu F = \operatorname{Im}(\psi^6) = \pi$ for $m = 6$. These are equalities for the exact solution, including the wall endpoints by continuity.

The field is analytic in D by elliptic regularity. Its zero Dirichlet data and the displayed analytic Neumann data on the analytic boundary give a local analytic continuation by the Cauchy–Kowalevski theorem. To identify it with F on the interior side, their difference has zero Cauchy data. Extend the difference by zero across the boundary; integration by parts shows that it solves Helmholtz distributionally there. Analytic elliptic regularity and vanishing on the exterior open set imply that the difference is zero. Compactness of the boundary gives an analytic continuation on a neighbourhood of $\bar{D}$.

The real coefficient and sine symmetries commute with the equation and the conformal change of variables. For $m = 1$, D is invariant under both coordinate reflections and F is odd in y . For $m = 6$, $\psi(e^{i\pi/6}w) = e^{i\pi/6}\psi(w)$ and F changes sign under the corresponding rotation. It is odd under conjugation, and hence under the Weyl reflections. In particular D is Weyl-invariant and F is Weyl anti-invariant.

7.2. The four-dimensional lift. Define

$$\Omega_4 = \{(x, X') : (x, |X'|) \in D\}, \quad u_4(x, X') = G(x, |X'|^2), \quad F(x, y) = yG(x, y^2).$$

The division is analytic locally at every axis point. Away from the axis it is ordinary division by y , so the local definitions agree. Equation (2.1) proves the Helmholtz equation, and the boundary product rule proves $u_4 = 0$, $\partial_\nu u_4 = 1$, extending to the axis by analyticity.

Convexity follows from planar convexity and reflection symmetry: for two points the norm of their transverse convex combination is at most the corresponding convex combination of transverse norms. The planar section contains every smaller nonnegative vertical coordinate at

the same horizontal coordinate. Strict convexity follows as well. If the meridian data of two different boundary points differ, their planar convex combination is interior. If they coincide but the transverse directions differ, strict triangle inequality decreases the transverse norm, again giving an interior point. At transverse norm zero identical meridian data give a single point, so this case has no exception.

For completeness the analytic boundary can be written as a radial sphere. The positive meridian radius $R_D(\theta)$ is analytic, even and π -periodic. Its exponentially convergent Fourier series uses only $\cos(2k\theta) = T_k(\cos 2\theta)$. On S^3 substitute $\cos 2\theta = \omega_1^2 - |\omega'|^2$. The resulting positive function $R(\omega)$ is analytic: the exponential coefficient bound makes the Chebyshev series converge on a complex neighbourhood of the compact range $[-1, 1]$. The radial map $\omega \mapsto R(\omega)\omega$ is injective, and its derivative has tangential projection $R(\omega)$ times the identity. It is therefore an analytic parametrization of the boundary by S^3 .

The symmetries force the centre of a hypothetical ball to be zero. Its meridian section would be a centred disc. A conformal map to that disc with $\psi(0) = 0$, $\psi'(0) > 0$ is linear, contrary to $c_3^* \neq 0$. Finally (5.16), on the positive meridian half-plane, gives opposite signs of u_4 . This proves Theorem 1.1.

7.3. The invariant lift in dimension fourteen. Set $\Omega_{14} = \text{Ad}(G_2)D$. Every orbit meets $\mathfrak{t}$ in a Weyl orbit, so $\Omega_{14} \cap \mathfrak{t} = D$ and the regular field is defined by $u_{14}|_{\mathfrak{t}} = F/\pi$. Analytic anti-invariance makes F vanish on each wall. Analytic division by a linear wall factor can be repeated: after division the remaining wall vanishing follows off the intersections and then by analytic continuation. Thus F/π is analytic and Weyl-invariant on the whole Cartan section, including the origin. The field scale and the normalization of π have been fixed throughout. Using the literal positive-root product instead would multiply the constant flux by a nonzero scalar, removable by scaling u_{14} .

We state the convexity result used here precisely. For a real semisimple Cartan decomposition $\mathfrak{g} = \mathfrak{t}_c \oplus \mathfrak{p}$, with maximal abelian section $\mathfrak{a} \subset \mathfrak{p}$, Kostant's theorem says $P_{\mathfrak{a}}(\text{Ad}(K)x) = \text{conv}(Wx)$ for $x \in \mathfrak{a}$. Lewis [10], Theorems 2.2 and 3.6, formulates its consequence: an $\text{Ad}(K)$ -invariant subset of $\mathfrak{p}$ is convex exactly when its section in $\mathfrak{a}$ is convex. The projection theorem is due to Kostant [9]. Apply this to the complexification of $\mathfrak{g}_2$ regarded as a real Lie algebra, with $\mathfrak{p} = i\mathfrak{k}$, $\mathfrak{a} = i\mathfrak{t}$ and compact group G_2 . Multiplication by i is an equivariant isometry after normalization of the Killing form. The convex Weyl-invariant section therefore gives convexity of Ω_{14} .

One can also see the set conclusion from the projection statement. The closed invariant lift of $\overline{D}$ equals the intersection of all adjoint conjugates of $\{X : P_{\mathfrak{t}}X \in \overline{D}\}$. The inclusion from left to right uses $\text{conv}(Wx) \subset \overline{D}$. For the reverse inclusion, conjugate X into $\mathfrak{t}$ and apply that particular projection. The intersection is convex. Its interior is $\text{Ad}(G_2)D$: the radial description below identifies interior and boundary. Thus the open domain is convex.

To justify the analytic boundary, let $R_D(\theta)$ be the positive analytic radial function of D . Weyl symmetry makes it an analytic function $f(\cos 6\theta)$. At the endpoints of $[-1, 1]$ this follows from the even analytic Taylor expansion at a reflection wall and the nonzero quadratic term of $\cos 6\theta$ there. Polynomial Chevalley restriction gives a real homogeneous invariant polynomial P_6 on $\mathfrak{k}$ restricting to $\text{Re } z^6$; see, for example, the restriction statement in [14]. Hence on an annulus around the unit sphere

$$R(X) = f\left(\frac{P_6(X)}{|X|^6}\right)$$

is analytic. Every unit direction is conjugate to a Cartan direction, so its argument lies in $[-1, 1]$ and R is positive. This yields an analytic radial boundary with nonzero normal, parametrized by S^{13} .

The regular orbit field satisfies Helmholtz by (2.3). Its extension as an orbit function is locally bounded since the analytic section quotient is locally bounded. The nonregular set has codimension at least three: a nonzero wall has one section parameter and an extra two-dimensional root plane in its centralizer, so its orbit saturation has dimension $1 + (14 - 4) = 11$; the origin has higher codimension. On compact subsets away from the origin these are finitely many smooth wall strata. An ϵ -tube cutoff around a stratum of codimension $c \geq 3$ has gradient and Hessian integrals $O(\epsilon^{c-1})$ and $O(\epsilon^{c-2})$. Testing Helmholtz against this cutoff, using boundedness of the field, makes all error terms tend to zero. A separate ball cutoff treats the origin. The equation consequently holds distributionally across the nonregular set. Elliptic analytic regularity gives the analytic extension. Apply the same argument to the planar continuation on a slightly larger invariant neighbourhood to obtain boundary regularity.

At regular boundary points $u_{14} = 0$ and $\partial_\nu u_{14} = 1$ by (2.4). The regular boundary points are dense, and the analytic field and boundary extend both equalities to all boundary points. The domain is bounded, and its Cartan section is strictly convex. An adjoint-invariant ball would have an adjoint-fixed centre; the simple Lie algebra has no nonzero fixed vector. Its section would therefore be a centred disc, contradicting $c_{13}^* \neq 0$. The two points of Section 6 have positive π and opposite field signs. This proves Theorem 1.2.

8. THE COMPUTER-ASSISTED INEQUALITIES

The verification directories contain the two self-contained Python programs, their transparent coefficient programs and exact data, supplementary numerical comparisons, and SHA-256 lists. All coefficient calculations use rational arithmetic or Arb interval arithmetic through `python-flint` 0.9.0. The recorded Python version is 3.10.19 on Linux x86-64. Arb calculations use one thread; the surrounding numerical libraries are also restricted to one thread. Binary64 files specify exact dyadic inputs.

The four-dimensional single file authenticates its payload and manifest, checks every finite and near-tail span, reconstructs and compares the finite inverse, and recomputes the residual, far-tail, nonlinear, rational-radius and geometric inequalities. Its complete mode additionally regenerates every bounded column file and compares the bytes. The fourteen-dimensional file authenticates its payload and individual files, checks finite and cross-column coverage, recomputes the inverse profile and far bounds, composes the all-column defect, regenerates both nonlinear coefficient files with $\theta_1 = 1$, transfers the centre, and checks the radius, boundary signs and geometry. Its complete mode regenerates every finite and near-column file.

The finite inverse and near columns do not themselves prove an infinite-dimensional norm bound. The support arguments, character identities, far-tail estimates, injectivity and reconstruction are proved in the text. The programs enclose the finite expressions in those estimates. The additional `check_inequalities.py` compares their typed interval outputs with the sharper numerical ceilings printed here. The finite matrix program enforces the ceiling $21 \cdot 10^{-35}$ used in the assembled estimate; the supplementary comparisons also check the sharper finite defect in (5.1).

The principal contraction bounds are:

Quantity	Dimension four	Dimension fourteen
r	1/12500	10^{-20}
Y	$< 3.038 \cdot 10^{-6}$	$< 1.063 \cdot 10^{-25}$
Z_0	< 0.926	< 0.49
Z_1	0	$< 2.4 \cdot 10^{-11}$
$Z_2(r)$	< 0.042881	$< 4 \cdot 10^{-14}$
$\ A\ $	< 2444	$< 10^5$

For location of the numerical comparisons, the finite inverse and residual in dimension four are in `verify_finite_inverse_r4.py`; its five column classes are in the three bounded-column families and `verify_far_source_bound_r4.py`, `verify_far_shape_bound_r4.py`. Its remaining inequalities are in `verify_nonlinear_r4.py`, `verify_radii_r4.py`, `verify_geometry_r4.py` and `verify_signchange_r4.py`. `verify_ball_linearization.py` also checks a Dirichlet-ball Bessel near-resonance and the quadratic and cubic moments of U_4 ; this is a consistency check on which no inequality of the proof depends. In dimension fourteen the finite matrix is checked by `verify_finite_block_rank2.py`; the tail and cross estimates are checked by the Schur, source-tail, cross-source, cross-shape, H and QD programs. The finite-input files and `verify_last_finite_column_r14.py` complete that column class. The nonlinear and centre-transfer calculations are in `nonlinear_majorant_rank2.py` and `verify_center_shift_r14.py`, and the final inequalities in `verify_radii_geometry_r14.py`.

From the corresponding verification directory run, sequentially,

```
OMP_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 MKL_NUM_THREADS=1 \
  nice -n 19 python3 -B verify_berenstein_r4.py
OMP_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 MKL_NUM_THREADS=1 \
  nice -n 19 python3 -B verify_r14.py
```

Each command ends with the line `PROVED` and exit code zero. For complete regeneration append `--recompute` to the four-dimensional command and `--recompute-all` to the fourteen-dimensional command. Run the additional numerical comparisons with `nice -n 19 python3 -B check_inequalities.py`. Its last line is `ALL_PAPER_INEQUALITIES_CHECKED`.

Recorded single-file wall times are 9.514 seconds and 34.589 seconds, respectively. Complete serial regeneration of the supplied mathematical programs and numerical data took 374.957 seconds and 518.356 seconds. Optimization with `-O` is rejected before mathematical verification. In each dimension a changed embedded numerical file was also rejected after its outer payload digest was recomputed, because the inner file manifest did not match.

The single-file and compressed-payload SHA-256 values are Dimension four, program:

```
e3479208ba7234e291cdde4d30e8e8c66ece07ae3bdade67daf9661f622a6337
```

Dimension four, payload:

```
77cebdd65d09b3f10ab95a17efb05af1f5b8add1837c1a5d6973b679ced1ba97
```

Dimension fourteen, program:

```
3ea6375202d31df994cb407603dc7d3c7cbee83587b95c91ab4fe71e8a3b5702
```

Dimension fourteen, payload:

```
a6229049ed6e91c4a690c1f0ed9ad49034272328751d70207cc5bda6829dc88e
```

Each directory's `SHA256SUMS` also lists every transparent verification file. A hash comparison authenticates the specified reproducible inputs; the analytic proof and the interval calculations establish the theorem.

9. REMARKS

The two solutions lie outside the sign-definite case of Serrin's theorem. In dimension fourteen the assertion of strict convexity is made for the Cartan section; the ambient-domain assertion is convexity. Dimensions six, eight and ten are not covered by these certificates. The two coefficient balls specify the exact constructions.

USE OF AI

The proofs, the programs and the text of this paper were produced with AI systems (GPT-6 Astra, GPT-6.1 Sol and Claude Opus 5.5) under the author's direction. They were checked by independent reviews with AI models and by independent replays of the interval-arithmetic certificates. The author takes responsibility for the content.

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Email address: mitsuha2021b@gmail.com